

Ephemeral Meta-Weights for Bandwidth-Aware Transformer Feed-Forward Networks

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Abstract

Transformer inference and training increasingly run into a weight-bandwidth limit: feed-forward network (FFN) projections are large, reused at every token, and repeatedly streamed from off-chip memory. We study MWG-EW, a neural architecture that replaces selected persistent FFN matrices with conditional low-rank meta-weights. A small generator produces rank- r descriptors for the current layer and token group; the target runtime consumes those descriptors with an associative $(XU_r)V_r$ schedule and a no-write-back lifecycle. In the transformed block, the dense FFN matrix is no longer stored or synchronized as a trainable object, while the present prototype measures an unfused associative path and still requires fused-kernel counter validation.

This paper makes three contributions. First, we define a generator-conditioned Transformer FFN that preserves differentiability while exposing rank, token reuse, and basis-bank size as architecture knobs. Second, we give an infrastructure contract for ephemeral descriptor execution and derive compute, bandwidth, memory, latency, and gradient-communication scaling laws. Third, we provide a distributed evaluation harness and report wall-clock latency, traffic estimates, communication volume, checkpoint-backed approximation quality, and small end-to-end perplexity gates on a patched Qwen2.5-1.5B-Instruct model. On an eight-accelerator ASI1 Ascend 910B2 prototype, the current implementation at $r = 128$ reduces estimated descriptor bytes by 24.9x and trainable FFN state by 18.4x, improves the measured training step by 1.542 ± 0.196 x, and reduces measured all-reduce latency by 5.4x. The fastest measured point, $r = 256$, reaches 1.572 ± 0.165 x training speedup; the lower-rank $r = 64$ point gives 49.8x estimated descriptor-byte reduction. A checkpoint-backed Qwen2.5-1.5B FFN distillation experiment shows that short low-rank distillation improves over static SVD but remains far from dense-output fidelity on early layers. A quality-gated ASI3 probe identifies layer 16 as the least brittle replacement candidate, while a two-layer patch that looked strong on a 4080-token diagnostic gate does not generalize to a more diverse text stress set. Direct LM-CE calibration improves layer 16 on the original held-out instruction slice, but a new manifest-driven broad validation still degrades GSM8K, MBPP, and Alpaca-tail suites, and a deployable router stays close to dense only when it patches a very small token fraction. The central claim is therefore a feasibility case and negative validation study rather than a final deployment claim: learned conditional generation, eligibility selection, and a no-write-back operand lifecycle can make parameter movement an architectural design variable, provided future training, fused kernels, and hardware counters close the current quality and traffic-evidence gaps.

1 Introduction

The standard Transformer block stores FFN projections as persistent dense matrices and streams them during every forward pass. For autoregressive decoding, the arithmetic per token can be small relative to the bytes fetched from high-bandwidth memory. For distributed training, the same persistent matrices induce equally persistent gradient synchronization traffic. This paper asks

whether a Transformer can improve this operating point by changing the lifecycle of FFN weights rather than only changing their datatype.

MWG-EW turns a subset of FFN weights into generated, short-lived descriptors. The descriptor is not a compressed checkpoint artifact; it is an operand created by a neural generator, consumed by a kernel, and then overwritten. The architecture is therefore both algorithmic and infrastructural: the learning rule determines which descriptors are useful, while the runtime determines whether the descriptors actually avoid off-chip traffic.

This distinction matters because many efficient-model methods still leave the same lifecycle in place: a parameter tensor is stored, loaded, differentiated, and synchronized as a durable object. MWG-EW studies the complementary question: what happens if a large projection is a transient computation product rather than a resident model object? If this view holds under full training and quality validation, it suggests a new axis for scaling Transformers: not only fewer parameters or fewer bits per parameter, but shorter parameter lifetimes.

The proposal is close in spirit to hypernetworks [3], low-rank adaptation [4], and IO-aware kernels such as FlashAttention [1], and it is complementary to attention-side KV-cache compression such as Multi-head Latent Attention [2]. Hypernetworks usually materialize generated weights as ordinary model tensors. LoRA stores low-rank factors as persistent adapter parameters. FlashAttention optimizes attention activation traffic, while MLA compresses attention keys and values during autoregressive decoding. MWG-EW instead treats generated FFN factors as volatile kernel operands.

Contributions. We make the following contributions:

- A Transformer FFN replacement in which a learned generator emits conditional rank- r descriptors for the up, gate, and down projections.
- A no-write-back descriptor lifecycle that connects the neural architecture to memory traffic and distributed synchronization.
- Scaling laws for parameter bytes, descriptor bytes, latency in the memory-bound regime, and all-reduce volume.
- A distributed benchmark that records latency, memory-traffic estimates, and communication evidence in machine-readable form.
- A negative static-SVD baseline and end-to-end patch tests showing why the method should be judged as conditional generation, eligibility selection, patch-aware calibration, and lifecycle control rather than ordinary low-rank compression.
- A manifest-driven broad-validation and deployable-router study showing that the current layer-16 checkpoint is not yet broadly quality preserving.

2 Architecture

2.1 Baseline Gated FFN

A gated Transformer FFN maps hidden states $X \in \mathbb{R}^{B \times S \times d}$ to

$$Y = (\phi(XW_{\text{gate}}) \odot XW_{\text{up}}) W_{\text{down}}, \quad (1)$$

where $W_{\text{up}}, W_{\text{gate}} \in \mathbb{R}^{d \times m}$, $W_{\text{down}} \in \mathbb{R}^{m \times d}$, and ϕ is typically SiLU. With fp16 weights, one block stores approximately $6dm$ bytes when m dominates d .

2.2 Ephemeral Meta-Weight FFN

MWG-EW replaces each projection matrix by a conditional descriptor pair:

$$(U_r^{(p)}, V_r^{(p)}) = G_\theta(c_t, \ell, p), \quad XW_p \approx (XU_r^{(p)})V_r^{(p)}, \quad (2)$$

where $p \in \{\text{up, gate, down}\}$, c_t is a token-group context, ℓ is the layer id, $U_r \in \mathbb{R}^{d_{\text{in}} \times r}$, and $V_r \in \mathbb{R}^{r \times d_{\text{out}}}$. The FFN becomes

$$\tilde{Y} = \left(\phi((XU_r^{(g)})V_r^{(g)}) \odot (XU_r^{(u)})V_r^{(u)} \right) U_r^{(d)}V_r^{(d)}. \quad (3)$$

We use a basis-bank generator:

$$\alpha = \text{softmax}(A_\ell h_\theta(c_t)), \quad (4)$$

$$U_r^{(p)} = \sum_{k=1}^K \alpha_k U_k^{(p)} + \Delta U_\theta^{(p)}(c_t, \ell), \quad (5)$$

$$V_r^{(p)} = \sum_{k=1}^K \alpha_k V_k^{(p)} + \Delta V_\theta^{(p)}(c_t, \ell). \quad (6)$$

The bank gives stable reusable directions; the residual head gives input-conditioned adaptation. Practical kernels may generate only rank-channel scales or tile-local residuals when full descriptor synthesis is too expensive.

2.3 Descriptor Lifecycle

The runtime contract is simple and testable:

1. Read token context c_t and layer id ℓ .
2. Generate descriptor factors (U_r, V_r) for each FFN projection.
3. Store descriptor tiles in an on-chip scratch region when the target accelerator exposes such memory; otherwise keep the same volatile lifecycle at the closest available cache level.
4. Compute each projection with the associative schedule $(XU_r)V_r$.
5. Release or overwrite each scratch tile before the next token group.
6. Synchronize gradients only for generator and basis-bank parameters.

The important distinction is not only low rank. A low-rank factor stored as a long-lived model tensor still creates persistent parameter traffic. The ephemeral contract allows a runtime to avoid writing generated descriptors back to off-chip memory and allows distributed training to synchronize only the generator-side state.

3 Analysis

3.1 Traffic Scaling

For a gated FFN with three projections and fp16 elements, dense weight traffic per block is

$$\mathcal{T}_{\text{dense}} = 6dm. \quad (7)$$

Ignoring the small generator context, rank- r descriptor traffic is

$$\mathcal{T}_{\text{mwg}} = 2r(d + m) + 2r(d + m) + 2r(m + d) = 6r(d + m). \quad (8)$$

The idealized traffic ratio is therefore

$$\Gamma(r) = \frac{\mathcal{T}_{\text{mwg}}}{\mathcal{T}_{\text{dense}}} = \frac{r(d + m)}{dm}. \quad (9)$$

At $d = 4096$, $m = 14336$, and $r = 128$, this gives $\Gamma = 0.0402$, or a 24.9x reduction in descriptor bytes.

Proposition 1 (Memory-bound speedup condition). *Assume per-block latency can be approximated by $L = L_0 + \kappa\mathcal{T}$ with fixed launch/compute term L_0 and bandwidth coefficient $\kappa > 0$. Then MWG-EW improves latency whenever $\mathcal{T}_{\text{mwg}} < \mathcal{T}_{\text{dense}}$, and the asymptotic speedup approaches $1/\Gamma(r)$ as L_0 becomes small relative to memory traffic.*

3.2 Gradient Communication

In data-parallel training, dense FFN gradients require all-reduce traffic proportional to $6dm$ bytes per block. MWG-EW synchronizes generator and descriptor-bank parameters, whose leading term is $6r(d + m)$ plus the small MLP state. The same ratio $\Gamma(r)$ controls the dominant gradient-byte reduction. This is orthogonal to optimizer sharding and tensor parallelism: those systems can still be used on the remaining generator parameters.

3.3 Computational Complexity

Let $T = BS$ be the number of tokens processed by one FFN call and let q be the token-group reuse factor for a generated descriptor. The dense gated FFN performs three large matrix products and has leading compute

$$\mathcal{C}_{\text{dense}} = \Theta(3Tdm) \quad (10)$$

multiply-accumulate operations. The associative MWG-EW path computes XU_r and then ZV_r for the up, gate, and down projections:

$$\mathcal{C}_{\text{mwg,assoc}} = \Theta(3Tr(d + m)). \quad (11)$$

If a basis bank with K entries is explicitly blended once per token group, the descriptor-generation term is

$$\mathcal{C}_{\text{gen}} = \Theta(3 \lceil T/q \rceil Kr(d + m) + \lceil T/q \rceil C_G), \quad (12)$$

where C_G is the small context-network cost. Kernels that generate only rank-channel scales or tile-local residuals replace the $Kr(d + m)$ term by the corresponding tile size. Thus the leading compute ratio, ignoring amortized generation, is again $\Gamma(r) = r(d + m)/(dm)$. The method is attractive when $r \ll \min(d, m)$ and q is large enough to amortize generation; it can be unattractive when descriptors are regenerated for very small token groups or when unfused factor matmuls lose accelerator utilization.

Table 1: Static SVD baseline on 18 pretrained FFN matrices. Low ranks are excellent for bytes but weak as fixed approximations, motivating learned conditional generation.

Rank r	Mean energy	Min energy	Mean cosine
32	7.17%	4.27%	0.266
64	11.98%	8.16%	0.345
128	20.12%	15.46%	0.448
256	33.79%	28.65%	0.582
512	55.61%	50.75%	0.748
1024	85.13%	82.42%	0.925

3.4 Working-Set Complexity

The persistent FFN parameter state is $\Theta(3dm)$ elements per layer. A fully materialized rank- r descriptor set is only $\Theta(3r(d+m))$ elements, and a tiled kernel can reduce the live scratch working set further to $\Theta(r(d_{\text{tile}} + m_{\text{tile}}))$. These are storage-footprint and traffic estimates. They are not claims that every descriptor or activation is resident in on-chip SRAM at once; the no-write-back contract says that generated descriptor tiles should be consumed at the closest available scratch/cache level and not committed as durable off-chip model weights.

3.5 Approximation

Let $\widetilde{W}_p = U_r^{(p)} V_r^{(p)}$ be the generated approximation of W_p . If $\|W_p - \widetilde{W}_p\|_F \leq \epsilon_p$, then for bounded activations and L_ϕ -Lipschitz nonlinearity,

$$\|Y - \widetilde{Y}\|_F \leq C_1 L_\phi \|X\|_F (\epsilon_{\text{gate}} + \epsilon_{\text{up}}) + C_2 \|X\|_F \epsilon_{\text{down}}, \quad (13)$$

where C_1 and C_2 depend on activation bounds inside the FFN. This connects rank selection to output distortion, while leaving room for end-to-end distillation to improve beyond static SVD.

3.6 Static Low-Rank Is Not the Claim

To separate MWG-EW from ordinary compression, we analyze a static truncated-SVD baseline on 18 FFN matrices from a Qwen-1.5B-family checkpoint. Table 1 reports mean retained singular-value energy and reconstruction cosine. At ranks that are attractive for bandwidth, static SVD preserves too little FFN structure to be a complete solution. This is evidence for the generator side of MWG-EW: the low-rank factors should be conditional operands, not merely fixed compressed weights.

4 Systems Realization

An implementation needs three pieces:

- **Generator placement.** The generator should run adjacent to the FFN kernel and emit descriptor tiles directly into scratch memory.
- **Associative execution.** The kernel computes XU_r followed by ZV_r , fusing activation and gate operations where possible.

- **Lifecycle enforcement.** Descriptor buffers are marked volatile; runtime instrumentation checks that descriptor write-back does not occur.

The benchmark in this folder measures the associative path and the distributed communication effect. A production kernel should additionally expose hardware counters for off-chip descriptor reads and writes.

5 Experimental Protocol

Execution setup. The large-scale reliability run uses the ASI1 Huanxin environment with eight visible Ascend 910B2 NPUs. We evaluate one, two, four, and eight data-parallel ranks by setting `ASCEND_RT_VISIBLE_DEVICES` to the corresponding device set and launching PyTorch distributed execution with HCCL. Each configuration is repeated three times. The raw artifact records platform metadata for reproducibility, but the method and results below are reported in terms of model geometry, rank, latency, state size, estimated traffic, and collective communication.

Geometry. The large settings use an 8B-scale FFN block with $d = 4096$, $m = 14336$, decode shape $B = 1$, $S = 128$, and training shape $B = 2$, $S = 512$. The reliability/scaling sweep reported below uses representative ranks $r \in \{64, 128, 256\}$.

Validation path. The benchmark uses a small smoke geometry $(d, m) = (512, 1408)$ with short iteration counts before running the 8B-scale geometry. This keeps the expensive distributed sweep separate from basic checks of output writing, tensor shapes, and collective communication.

Metrics. We report forward latency, backward latency, throughput, trainable parameter bytes, estimated descriptor traffic, dense-to-MWG-EW speedup, all-reduce latency, and all-reduce bytes.

Checkpoint quality and end-to-end probes. To avoid treating systems speed as a proxy for model quality, we separately distill the FFN output of layer 0 from a local Qwen2.5-1.5B-Instruct checkpoint stored in the ASI1 file system. The teacher uses the checkpoint `gate_proj`, `up_proj`, and `down_proj` matrices with $d = 1536$ and $m = 8960$. We compare static truncated SVD, a trainable persistent low-rank FFN, and the same low-rank factors with the MWG-EW rank-scale generator. The reported run uses eight NPUs, fp32 training for numerical stability, ranks 64 and 128, 16 short distillation steps, and four evaluation batches. We then run a stronger ASI3 quality path on the same checkpoint: text-activation distillation, expert-residual MWG-EW students, saved student checkpoints, and direct replacement of selected Qwen MLP modules followed by causal-language-model perplexity evaluation. This second probe is still small, but it is end-to-end: the patched model is evaluated through the full Transformer, not only by block MSE.

Validation controls. After auditing an invalid recovery artifact that evaluated on the tiny default calibration text stream, all held-out-named recovery and broad-validation launchers now require explicit text inputs. The ASI3 recovery path refuses `DEFAULT_TEXTS` fallback through a `--require-texts` guard, and a manifest-driven broad-validation harness rejects empty or undersized corpora before launching PPL evaluation. These controls are procedural rather than new quality evidence, but they are important for reproducibility: held-out results below are only comparable when the text source and token count are explicit.

Deployable selective routing protocol. The next algorithmic validation step is a non-oracle dense-fallback policy for the best current single-layer patch. We added a router evaluation harness that uses one explicit text split to label examples by dense-vs-patched loss delta, fits a small ridge regressor from features available before the target FFN executes, and evaluates thresholded patch fractions on a disjoint held-out split. The features are summary statistics of the layer-16 pre-FFN hidden state and sequence occupancy, not evaluation-set dense-vs-patched losses or post-hoc perplexity comparisons. Its report includes always-dense, always-patched, deployable routed, and oracle-frontier rows. This protocol is intended to test whether MWG-EW can become a selective runtime optimization rather than an indiscriminate layer replacement; it does not add a positive claim until broad held-out corpus files and multiple seeds have been run.

Artifact trail. The reported ASI1 systems run is stored under the historical artifact name `ai3_reliability_scaling_20260518T072507Z`. It contains 12 raw JSON benchmark artifacts: three repeats each for world sizes 1, 2, 4, and 8, plus an aggregate JSON/Markdown summary. The local repository keeps the pulled summary, status file, and log under `results/pulled_as11/`; the same experiment code and paper artifacts are mirrored to the project S3 folder.

6 Numerical Evidence

Evidence status. The results below were measured on 2026-05-18 on ASI1 using all eight visible NPUs for the largest setting, `torch.distributed.run`, HCCL, and fp16 tensors. The benchmark uses the 8B-scale FFN geometry $(d, m) = (4096, 14336)$, decode shape $(1, 128, 4096)$, and training shape $(2, 512, 4096)$. Each row reports the mean and standard deviation over three independent launches. Each launch used three warmup iterations, ten forward timing iterations, three training iterations, and five all-reduce iterations. These measurements establish systems behavior for the associative low-rank path and distributed communication. They do not yet establish final end-to-end language-model quality.

Interpretation. The most stable eight-rank point is $r = 128$: it reduces descriptor traffic from 336.0 MiB to 13.5 MiB, reduces trainable FFN state from 336.0 MiB to 18.254 MiB after generator overhead, improves mean training latency from 6.4874 ± 0.8606 ms to 4.2047 ± 0.0299 ms, and increases measured training throughput from 1.279M to 1.948M tokens/s across eight ranks. The fastest eight-rank point in this suite is $r = 256$, reaching 1.572 ± 0.165 x training speedup and 1.992M tokens/s, while still reducing descriptor traffic by 12.4x and trainable FFN state by 10.3x. The lower-rank $r = 64$ point gives the strongest compression, with 49.8x descriptor-traffic reduction and 30.2x trainable-state reduction. Across 2, 4, and 8 ranks, the all-reduce proxy drops from dense latencies of 19.66, 11.49, and 6.91 ms to low-rank latencies near 1–2.5 ms, confirming that the trainable-state reduction translates into a distributed communication benefit. The forward-only path remains sensitive to unfused factor-matmul overhead; fused descriptor generation, on-chip factor consumption, and no descriptor write-back remain the systems requirements that separate MWG-EW from an ordinary low-rank module.

Table 4 is deliberately more conservative. On a real Qwen2.5-1.5B FFN layer, static SVD at $r = 64$ and $r = 128$ is a weak approximation despite strong byte savings. Short distillation improves relative MSE and cosine, but the current rank-scale MWG-EW generator is almost tied with the persistent low-rank baseline. This result prevents an overclaim: the present prototype establishes the communication and latency opportunity, while future work must use stronger conditional generators, longer distillation, or end-to-end training to convert that systems opportunity into language quality.

Table 2: Reliability and scaling of 8B-scale FFN training latency. Each cell is mean $\pm$ std over three launches on ASI1. The current prototype uses unfused associative execution, not yet a fused no-write-back kernel.

World	Rank	Train ms	Train speedup	Train tok/s
1	–	4.9742 $\pm$ 0.0239	1.000x	0.206M
1	64	3.3785 $\pm$ 0.1928	1.475 $\pm$ 0.075x	0.304M
1	128	3.2545 $\pm$ 0.1694	1.531 $\pm$ 0.071x	0.315M
1	256	3.2879 $\pm$ 0.1327	1.514 $\pm$ 0.055x	0.312M
2	–	5.8624 $\pm$ 0.7332	1.000x	0.353M
2	64	3.9907 $\pm$ 0.1104	1.472 $\pm$ 0.211x	0.513M
2	128	4.7966 $\pm$ 1.5295	1.268 $\pm$ 0.214x	0.453M
2	256	3.8896 $\pm$ 0.0662	1.507 $\pm$ 0.181x	0.527M
4	–	6.1351 $\pm$ 1.1496	1.000x	0.682M
4	64	4.1078 $\pm$ 0.1168	1.497 $\pm$ 0.303x	0.998M
4	128	4.4547 $\pm$ 0.6583	1.412 $\pm$ 0.423x	0.932M
4	256	4.1109 $\pm$ 0.0797	1.496 $\pm$ 0.312x	0.997M
8	–	6.4874 $\pm$ 0.8606	1.000x	1.279M
8	64	4.8044 $\pm$ 1.1034	1.373 $\pm$ 0.182x	1.760M
8	128	4.2047 $\pm$ 0.0299	1.542 $\pm$ 0.196x	1.948M
8	256	4.1222 $\pm$ 0.2432	1.572 $\pm$ 0.165x	1.992M

Table 3: State size, descriptor traffic, and measured all-reduce latency for the same ASI1 reliability run. AllReduce latency cells are mean ms over three launches for the corresponding world size.

Rank	Params MiB	Descriptor MiB	Traffic red.	AR 2	AR 4	AR 8
–	336.000	336.000	1.000x	19.6627	11.4896	6.9101
64	11.127	6.750	49.778x	1.2599	1.1655	1.1549
128	18.254	13.500	24.889x	1.6772	1.4632	1.2786
256	32.507	27.000	12.444x	2.5398	1.8239	1.4182

Tables 5, 6, 7, 8, 9, 10, 11, 12, and 13 show the resulting pivot, while Table 14 records the current runtime instrumentation boundary. Early-layer replacement is not credible: even rank-384 experiments left layer 0 with unacceptable perplexity damage, and replacing layers 0, 4, 8, and 12 at rank 128 collapses perplexity. The usable regime is instead quality-gated and patch-aware, but the latest ASI3 stress test also shows how easy it is to overfit a small gate. Joint logit calibration of layers 8 and 16 reaches a 0.982x PPL ratio on the 4080-token repeated-text gate, but degrades to 1.357x on the diverse64 stress set. The more credible current policy is therefore stricter: keep early layers, layer 4, layer 8, layer 24, and multi-layer compositions dense unless they pass held-out gates, and patch only the safest eligible layer. In these experiments, layer 16 is the best candidate: rank 256 gives a 1.105x PPL ratio on diverse64, while a rank-384 retraining on diverse activation data gives 1.061x on the held-out split. However, the same layer-16 policy is not yet robust: on a larger Belle-style instruction slice, rank 384 has a 1.287x PPL ratio before logit calibration. End-to-end logit-MSE calibration improves that external-corpus gate monotonically in the current sweep, reaching 1.205x after 400 calibration steps and 1.189x after longer gentle continuation. Direct language-model cross-entropy (LM-CE) calibration is the stronger current variant: it reduces the Belle held-out

Table 4: Checkpoint-backed FFN approximation on ASI1 using Qwen2.5-1.5B layer 0. Lower relative MSE and higher cosine are better. The short eight-NPU distillation run improves over static SVD but shows that the current rank-scale generator does not yet recover dense FFN fidelity at bandwidth- attractive ranks.

Method	Rank	Rel. MSE	Cosine	Params MiB	Traffic red.
Static SVD	64	0.989094	0.133395	–	20.488x
Persistent low-rank	64	0.966727	0.225573	9.096	20.488x
MWG-EW rank-scale	64	0.966702	0.225632	9.096	20.488x
Static SVD	128	0.971749	0.187598	–	10.244x
Persistent low-rank	128	0.942873	0.257047	17.066	10.244x
MWG-EW rank-scale	128	0.942804	0.257052	17.066	10.244x

Table 5: Diagnostic end-to-end Qwen2.5-1.5B perplexity gates for patched MWG-EW FFN layers on ASI3. Lower PPL ratio is better. The 1020–4080-token rows are not validation claims; they show why early-layer and multi-layer policies must be rejected or re-tested on held-out text.

Patch policy	Rank	Tokens	Dense PPL	Patched PPL / ratio
Layers 0,4,8,12, independent	128	1020	314.44	2.18M / 6948x
Layers 8,12,16,20, independent	128	1020	314.44	590.56 / 1.88x
Layers 8,12,16,20, independent	256	1020	314.44	662.34 / 2.11x
Layers 8,16, independent	256	4080	314.44	360.50 / 1.146x
Layers 8,16, joint logit-calibrated	256	4080	314.44	308.76 / 0.982x

ratio to 1.180x and improves the small off-distribution checks to 1.087x on diverse64 and 1.040x on heldout12. A same-rank persistent low-rank baseline trained on the same layer-16 activation cache reaches 1.373x on the Belle held-out gate, so the calibrated MWG variant is better than this simple durable low-rank control in the present setting. However, the remaining degradation is still material, the diverse checks are small, and the result is from one checkpoint and one split. Table 7 is the decisive current counterweight: when the same layer-16 checkpoint is evaluated on explicit GSM8K, MBPP, and Alpaca-tail held-out files, all three suites degrade, with a 1.246x token-weighted aggregate PPL ratio. A same-rank persistent low-rank baseline trained from explicit router-train activations also fails the broad manifest, with a 1.279x token-weighted ratio in its separate launch; adding direct LM-CE calibration to that persistent baseline worsens the broad aggregate to 1.310x over the 91,352-token manifest. Thus the calibrated MWG-EW variant is not merely losing to a trivial durable low-rank baseline, but neither method is close to broad quality preservation. Tables 8 and 13 show that dense fallback can bound quality loss only by patching very few tokens. The stricter token-level router is trained on token CE deltas and then evaluated with actual mixed dense/patch forward passes. It exposes a large split shift: on router-train the patch is favorable (“patched” PPL ratio 0.437x over 30,053 tokens), but on disjoint router-eval it is unfavorable (2.062x over 35,118 tokens). Mixed forwarding keeps ratios near 1.01–1.05x only while patching about 0.4–3.6% of tokens, then degrades to 1.166x at 9.5% patching and 1.601x at 24.8% patching. However, Table 9 shows that this result is split-sensitive within the router distribution: when the two router corpora are pooled and repartitioned into three explicit train/eval folds, mean mixed-forward ratios fall below dense, reaching 0.946x at 6.9% actual patching and 0.922x at 17.7% patching. This is an encouraging signal for risk-aware routing, but it does not by

Table 6: Held-out and stress-test Qwen2.5-1.5B perplexity gates for patched MWG-EW FFN layers on ASI3. Lower PPL ratio is better. The diverse64 suite is a small mixed-domain stress set; heldout12 is a split held out from diverse activation capture; Belle256 is a larger instruction-style validation slice. The uncalibrated Belle256 rows use the first 50,838-token slice, while the calibrated and persistent-baseline rows use the 51,267-token held-out slice reserved for post-calibration evaluation.

Patch policy	Rank	Tokens	Dense PPL	Patched PPL / ratio
Layers 8,16, joint logit-calibrated, diverse64	256	5469	62.69	85.09 / 1.357x
Layer 16 only, diverse64	256	5469	62.69	69.24 / 1.105x
Layer 16 only, diverse64	384	5469	62.69	69.68 / 1.112x
Layer 16 only, heldout12	384	2560	58.14	61.71 / 1.061x
Layer 16 only, Belle256	256	50838	8.37	11.77 / 1.406x
Layer 16 only, Belle256	384	50838	8.37	10.77 / 1.287x
Layer 16 persistent low-rank, Belle256	384	51267	8.44	11.59 / 1.373x
Layer 16 only, Belle256, logit-MSE calibrated	384	51267	8.44	10.03 / 1.189x
Layer 16 only, Belle256, LM-CE calibrated	384	51267	8.44	9.96 / 1.180x
Layer 16 only, diverse64, LM-CE calibrated	384	2736	62.37	67.80 / 1.087x
Layer 16 only, heldout12, LM-CE calibrated	384	1280	58.12	60.48 / 1.040x

itself overturn the negative broad manifest because the folds are resamples of the router corpus rather than independent GSM8K, MBPP, or Alpaca-style benchmarks. Table 10 is the stricter check: for each broad suite, exact eval-line overlaps are removed from the router train pool before fitting the token router. Always-patched remains negative on all suites (1.324x on GSM8K, 1.113x on MBPP, and 1.110x on Alpaca-tail), so the patch itself is not broadly quality preserving. Selective routing is better: the broad token-weighted frontier reaches 0.992x at 2.1% actual patching and 0.987x at 8.5% actual patching, before degrading to 1.014x at 19.8% patching. Table 11 repeats this leakage-clean broad protocol over three explicit split seeds using 75% of each overlap-clean train pool. The bounded selective result repeats: target 0.05, 0.10, and 0.25 are below dense on every seed, with mean token-weighted ratios of 0.998x, 0.992x, and 0.986x, respectively. The high-patching target 0.50 row is unstable, with seed-level ratios spanning 0.994x to 1.031x. This supports a bounded selective-routing claim rather than an always-patched deployment claim. Table 12 is a stricter distribution-robust check: for each target broad suite, the router also trains on the two non-target broad suites but removes exact target-suite eval lines. The bounded signal survives at target 0.05 and 0.10 on every seed, with mean ratios of 0.997x and 0.993x. Target 0.25 is below dense on average but has one near-neutral/slightly-worse seed, and target 0.50 degrades on every seed. The example-level routed frontier similarly keeps ratios near 1.01–1.03x at 1.7–3.5% actual token patching, reaches 1.141x by 14.5% patching, and always-patched evaluation degrades to 2.063x on the router-eval split. The oracle frontier is better at the same nominal target fractions but is not deployable because it uses evaluation losses. At the present stage this is a reproducible feasibility signal plus a bounded selective-router result, not yet a top-journal-quality deployment claim; stronger training, stronger baselines, broader independent benchmark families, and hardware-counter or fused-kernel evidence remain required. The runtime probe in Table 14 sharpens that last limitation: even on an ASI3 host

Table 7: Manifest-driven broad held-out validation for the ASI3 layer-16 rank-384 LM-CE checkpoint and same-rank persistent low-rank baselines. These suites use explicit held-out files and reject empty or undersized text inputs. Lower PPL ratio is better. The baseline was launched separately and has different evaluated token counts, so it is a broad negative control rather than a token-identical paired comparison.

Policy	Suite	Tokens	Dense PPL	Patched PPL	Ratio
MWG LM-CE	GSM8K test	44,708	3.4941	4.7481	1.3589x
MWG LM-CE	MBPP test	13,301	3.9877	4.5276	1.1354x
MWG LM-CE	Alpaca-tail	33,343	3.8818	4.4249	1.1399x
MWG aggregate	–	91,352	–	–	1.2464x
Persistent LR	GSM8K test	31,329	4.7059	6.0889	1.2939x
Persistent LR	MBPP test	12,850	4.1360	5.3876	1.3026x
Persistent LR	Alpaca-tail	22,694	4.2148	5.2535	1.2464x
Persistent aggregate	–	66,873	–	–	1.2795x
Persist+LMCE	GSM8K test	44,708	3.4941	5.0680	1.4504x
Persist+LMCE	MBPP test	13,301	3.9877	4.1225	1.0338x
Persist+LMCE	Alpaca-tail	33,343	3.8818	4.7813	1.2317x
Persist+LMCE agg.	–	91,352	–	–	1.3099x

with visible Ascend 910B2 NPUs and available `npu-smi/msprof`, the current unfused associative prototype appears as ordinary matmul-like operators, doubles the matmul-like op count in this microprobe, and is slower than dense. It is useful negative fused-kernel evidence and tool-availability evidence, but it is not yet an off-chip traffic counter validation.

7 Discussion

MWG-EW is most useful when FFN weight traffic dominates and when the target hardware has enough arithmetic headroom to spend extra low-rank multiplies or generator work. It is less attractive when the model is already compute-bound, when rank must approach full dimension, or when the generator is placed so far from the FFN kernel that descriptor traffic reappears.

The architecture also changes the meaning of compression. The goal is not only to reduce checkpoint size, but to change where parameters live during execution. That distinction is why the method belongs partly to neural architecture design and partly to model-runtime infrastructure.

Relation to KV-cache compression and MLA. The KV-cache memory used during autoregressive decoding is a different object from the FFN weight traffic studied here. A standard attention layer stores past keys and values with leading size $\Theta(2Ln_hd_h)$ elements for sequence length L , head count n_h , and head dimension d_h ; in practical long-context serving this footprint is usually an off-chip memory allocation, although recently accessed tiles may pass through on-chip caches during an attention kernel. MLA reduces this attention-side storage by caching a latent key-value vector and reconstructing or absorbing key/value projections at attention time [2]. MWG-EW is orthogonal: it targets FFN projection weights, does not by itself compress the attention KV cache, and uses low-rank generation to change the lifecycle of weight operands rather than the

Table 8: Token-level dense-fallback router validation on disjoint router-train/router-eval text. The router is trained on token CE deltas and then evaluated with actual mixed dense/patch forward passes, not offline oracle loss mixing. Lower PPL ratio is better.

Target patch frac.	Actual token patch frac.	PPL ratio
0.01	0.0038	1.0132x
0.03	0.0124	1.0249x
0.05	0.0193	1.0360x
0.10	0.0363	1.0508x
0.25	0.0947	1.1665x
0.50	0.2485	1.6013x

Table 9: Three-fold token-level router cross-validation over the pooled router corpus. The 512 unique router lines are repartitioned into explicit 256/256 train/eval folds, then evaluated with actual mixed dense/patch forward passes. This tests split sensitivity within the router distribution; it is not a replacement for independent broad benchmarks. Lower PPL ratio is better.

Target patch frac.	Mean actual patch frac.	Mean ratio	Min ratio	Max ratio
0.01	0.0009	0.9989x	0.9955x	1.0009x
0.03	0.0038	0.9940x	0.9870x	1.0009x
0.05	0.0075	0.9896x	0.9796x	1.0007x
0.10	0.0193	0.9787x	0.9644x	0.9974x
0.25	0.0695	0.9462x	0.9282x	0.9618x
0.50	0.1766	0.9218x	0.9137x	0.9373x

representation of past-token activations.

Current limitations. Three gaps are intentionally left visible. First, the end-to-end quality evidence is still negative outside the narrowest gates: it covers a small Qwen2.5-1.5B text gate, a stricter single-layer eligibility policy, one larger instruction-style stress test that still shows material degradation, and a new broad held-out validation where the same layer-16 checkpoint degrades GSM8K, MBPP, and Alpaca-tail. The added manifest harness prevents default-text self-evaluation, and the deployable-router harness separates train-time risk labels from held-out routing decisions, but the current router preserves dense quality only at very small patch-token fractions. Second, descriptor lifecycle claims should be confirmed with low-level hardware performance counters for off-chip read and write transactions, not only with software-side theoretical byte estimates. Third, the current generator and associative path are deliberately small and benchmarkable; stronger generators and fused kernels may improve both quality and forward latency, but must be measured against their own placement and traffic costs.

8 Conclusion

This paper reframes Transformer FFN weights as generated ephemeral operands. By replacing persistent dense matrices with conditional rank- r descriptors and an associative execution path, MWG-EW reduces both memory traffic and distributed gradient communication in an eight-rank 8B-scale FFN systems benchmark. A checkpoint-backed Qwen2.5-1.5B probe adds the missing

Table 10: Leakage-clean broad token-router validation. For each broad suite, the router train pool removes exact lines overlapping that suite’s eval file before training. Values aggregate GSM8K, MBPP, and Alpaca-tail using token-weighted losses. Lower PPL ratio is better.

Target patch frac.	Actual patch frac.	Token-weighted ratio	Min suite	Max suite
0.01	0.0005	1.0010x	1.0001x	1.0050x
0.03	0.0019	1.0008x	1.0005x	1.0025x
0.05	0.0055	0.9989x	0.9970x	1.0004x
0.10	0.0208	0.9916x	0.9648x	0.9996x
0.25	0.0849	0.9868x	0.9197x	1.0038x
0.50	0.1975	1.0140x	0.8587x	1.0498x

Table 11: Three-seed leakage-clean broad token-router sweep. Each seed uses 75% of the overlap-clean router train candidates for each broad suite, then evaluates actual mixed dense/patch forwards on GSM8K, MBPP, and Alpaca-tail. Values report the mean across seeds; min and max are seed-level token-weighted broad ratios. Lower PPL ratio is better.

Target patch frac.	Mean actual patch frac.	Mean ratio	Min seed	Max seed
0.01	0.0007	1.0011x	1.0008x	1.0013x
0.03	0.0021	0.9998x	0.9980x	1.0013x
0.05	0.0049	0.9977x	0.9941x	0.9997x
0.10	0.0182	0.9920x	0.9890x	0.9952x
0.25	0.0852	0.9864x	0.9820x	0.9929x
0.50	0.2168	1.0164x	0.9943x	1.0312x

negative control: static and short-distilled low-rank FFNs do not recover dense quality when applied indiscriminately, especially in early layers or brittle multi-layer compositions. The stronger result is a stricter quality-gated policy: layer 16 alone is the best current candidate on small diverse gates, whereas a two-layer calibrated patch overfits the original small gate and the layer-16 patch still degrades on a larger instruction-style corpus, although LM-CE calibration reduces that degradation and improves the small diverse checks. The accompanying benchmark is therefore best read as a reproducible systems case for ephemeral operand lifecycles plus an initial algorithmic recipe and negative evidence for quality-preserving layer eligibility. Broad dataset validation now shows that the present checkpoint is insufficient; stronger training, stronger routing, and fused-kernel counter evidence are the next experimental steps.

References

- [1] Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Rudra, and Christopher Ré. FlashAttention: Fast and memory-efficient exact attention with IO-awareness. In *Advances in Neural Information Processing Systems*, 2022.
- [2] DeepSeek-AI. DeepSeek-V2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024.
- [3] David Ha, Andrew Dai, and Quoc V. Le. Hypernetworks. In *International Conference on Learning Representations*, 2017.

Table 12: Leave-suite-out broad auxiliary token-router sweep. For each target broad suite, router training uses the explicit router texts plus the other two broad suites, with exact target-suite eval lines removed. Values report the mean across three split seeds; min and max are seed-level token-weighted broad ratios. Lower PPL ratio is better.

Target patch frac.	Mean actual patch frac.	Mean ratio	Min seed	Max seed
0.01	0.0043	1.0007x	1.0004x	1.0009x
0.03	0.0166	0.9989x	0.9980x	1.0004x
0.05	0.0284	0.9975x	0.9964x	0.9996x
0.10	0.0565	0.9926x	0.9914x	0.9943x
0.25	0.1380	0.9943x	0.9889x	1.0004x
0.50	0.2806	1.0274x	1.0076x	1.0488x

Table 13: Deployable dense-fallback router validation on disjoint router train/eval text. Values are averages over three launched ASI3 runs, which are effectively identical because the current ridge router is deterministic for these inputs. Lower PPL ratio is better.

Policy	Target patch frac.	Actual token patch frac.	PPL ratio
Routed frontier	0.10	0.0169	1.0129x
Routed frontier	0.25	0.0345	1.0273x
Routed frontier	0.50	0.1453	1.1411x
Always patched	1.00	1.0000	2.0625x
Oracle frontier	0.10	0.1028	1.0139x
Oracle frontier	0.25	0.2517	1.0624x
Oracle frontier	0.50	0.5027	1.2000x

- [4] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

Table 14: ASI3 runtime profiler probe for the current unfused prototype. The environment reported 8 visible Ascend 910B2 NPUs and available `npu-smi/msprof`, but this PyTorch-profiler capture exposed CPU activities only. It is therefore operator-level evidence, not full off-chip hardware-counter validation. Lower time is better.

Case	Descriptor MiB	Mean ms	Matmul-like ops	Fused-like op
Dense	64.50	0.3391	27	no
MWG-EW $r = 128$	5.53	0.7638	54	no
MWG-EW $r = 256$	11.06	0.7559	54	no